

数学基础 (I) 特征值eigenvalue 与特征向量 eigenvector

在数学上, 特别是线性代数中, 对于一个给定的**线性变换A**, 它的特征向量v经过这个线性变换的作用之后, **得到的新向量仍然与原来的v保持在同一条直线上。但其长度或方向也许会改变。**即

$$Av = \lambda v$$

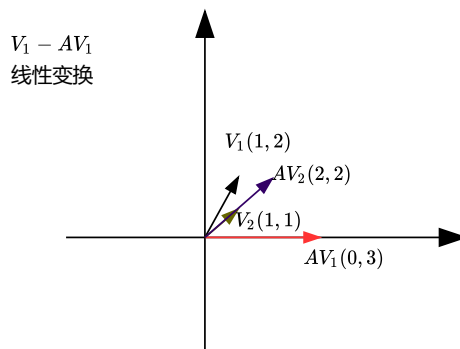
其中 λ 为标量, 即特征向量的长度在该线性变换下缩放的比例, 称其特征值。

线性变化

$$A = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \quad V_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$AV_1 = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 1 \times 2 \\ 4 \times 1 + (-2) \times 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad AV_2 = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2V_2$$



AV_2 和 V_2 在一条直线上

$AV_2 = 2V_2$ → 特征向量
λ, 特征值

求解特征值和特征向量

$$AV = \lambda V$$

$$AV - \lambda V = 0$$

$$(A - \lambda I)V = 0$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 \\ 4 & -2 - \lambda \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1 - \lambda & 1 \\ 4 & -2 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(-2 - \lambda) - 1 \times 4 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$(\lambda - 2)(\lambda + 3) = 0$$

特征值

$$\lambda_1 = 2$$

$$\lambda_2 = -3$$

要有非零解 ↓

$$|A - \lambda I| = 0$$

当 $\lambda_1 = 2$ 时

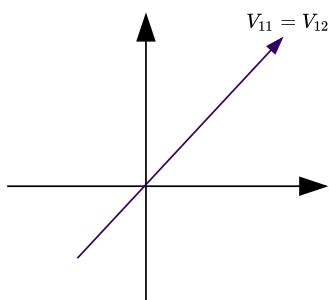
$$\begin{bmatrix} 1 - 2 & 1 \\ 4 & -2 - 2 \end{bmatrix} V_1 = 0$$

$$\begin{bmatrix} -1 & 1 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} V_{11} \\ V_{12} \end{bmatrix} = 0$$

$$\begin{aligned} -V_{11} + V_{12} &= 0 \\ 4V_{11} - 4V_{12} &= 0 \end{aligned}$$

$$V_{11} = V_{12}$$

可以任意取一组解 例如 $V_{11} = 1$
 $V_{12} = 1$ 特征向量 $V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$



同理, 当 $\lambda_2 = -3$

$$\begin{bmatrix} 1 + 3 & 1 \\ 4 & -2 + 3 \end{bmatrix} \begin{bmatrix} V_{21} \\ V_{22} \end{bmatrix} = 0$$

$$4V_{21} + V_{22} = 0$$

可取

$$\begin{aligned} V_{21} &= 1 \\ V_{22} &= -4 \end{aligned}$$

$$V_2 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

结果

$$A = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix}$$

$$\lambda_1 = 2 \\ \lambda_2 = -3$$

$$V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad V_2 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

应用：化对角矩阵，解耦，decouple

设一个过渡矩阵 $P = [V_1 \ V_2]$ P : coordinate transformation matrix

$$\begin{aligned} AP &= A[V_1 \ V_2] = A \begin{bmatrix} V_{11} & V_{21} \\ V_{12} & V_{22} \end{bmatrix} = \begin{bmatrix} A \begin{bmatrix} V_{11} \\ V_{12} \end{bmatrix} & A \begin{bmatrix} V_{21} \\ V_{22} \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 \begin{bmatrix} V_{11} \\ V_{12} \end{bmatrix} & \lambda_2 \begin{bmatrix} V_{21} \\ V_{22} \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 V_{11} & \lambda_2 V_{21} \\ \lambda_1 V_{12} & \lambda_2 V_{22} \end{bmatrix} \\ &= \begin{bmatrix} V_{11} & V_{21} \\ V_{12} & V_{22} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \\ &\quad \downarrow \quad \downarrow \\ &\quad P \quad \Lambda \quad \text{对角矩阵} \end{aligned}$$

$AP = P\Lambda$

$P^{-1}AP = P^{-1}P\Lambda$

$P^{-1}AP = \Lambda$

微分方程组，state-space

$$\frac{dx_1}{dt} = x_1 + x_2 \quad \Rightarrow \quad \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{dx_2}{dt} = 4x_1 - 2x_2$$

$$\dot{x} = Ax$$

$$P\dot{y} = APy$$

$$P^{-1}P\dot{y} = P^{-1}APy$$

解耦
非常漂亮的公式

$$\dot{y} = \Lambda y$$

$$\text{令 } x = Py$$

$$\dot{x} = P\dot{y}$$

$$Ax = APy$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$$

使用特征值和特征向量的方式来解微分方程组

$$\dot{y} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} y$$

$$\dot{y}_1 = 2y_1 \\ \dot{y}_2 = -3y_2$$

$$y_1 = C_1 e^{2t} \\ y_2 = C_2 e^{-3t}$$

C_1, C_2 是常数

$$x = Py = \begin{bmatrix} 1 & 1 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} C_1 e^{2t} \\ C_2 e^{-3t} \end{bmatrix} = \begin{bmatrix} C_1 e^{2t} + C_2 e^{-3t} \\ C_1 e^{2t} - 4C_2 e^{-3t} \end{bmatrix}$$

Summary

① $Av = \lambda v$ 在一条直线上

② 特征值、特征向量的求解方法 $|A - \lambda I| = 0$ $(A - \lambda I)V = 0$

③ $P^{-1}AP = \Lambda$ $P = [V_1, V_2, \dots]$ $\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \dots \end{bmatrix}$

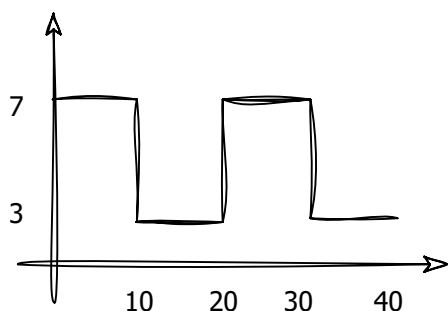
④ $\dot{x} = Ax$ 令 $x = Py$ $\dot{y} = \Lambda y$

很多的时候，我们并不一定要解出微分方程，而是通过判断特征值的符号以及性质来判断系统的稳定性和表现

傅里叶级数与变换

Part I 三角函数的正交性

分享知乎文章: [傅里叶分析之掐死教程 \(完整版\) 更新于2014.06.06](#)



$F(t) = ?$

周期函数, 如何展开为傅里叶级数

展开后的傅里叶级数, 每一个参数的意义是怎么来的

三角函数系: 集合

取两个不同的, 乘积在 $[-\pi, \pi]$, 定积分为0

$$\{0, 1, \sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin nx, \cos nx, \dots\}$$

$\sin 0x$ $\cos 0x$ $\sin nx$ $\cos nx$ $n = 0, 1, 2, 3, \dots$

正交: 上面集合任取两个不同的项

$$\int_{-\pi}^{\pi} \sin nx \cos mx dx = 0 \quad n \neq m$$

正交
垂直

$$\int_{-\pi}^{\pi} \cos nx \cos mx dx = 0 \quad n \neq m$$

证明:

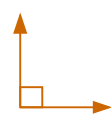
$$\int_{-\pi}^{\pi} \cos 0x \sin x dx = \int_{-\pi}^{\pi} \sin x dx = 0$$

$$\begin{aligned} & \int_{-\pi}^{\pi} \cos nx \cos mx dx \quad \text{积化和差公式} \\ &= \frac{1}{2} [\cos(n-m)x + \cos(n+m)x] \\ &= \frac{1}{2} \left[\int_{-\pi}^{\pi} \cos(n-m)x dx + \int_{-\pi}^{\pi} \cos(n+m)x dx \right] \\ &= \frac{1}{2} \left[\frac{1}{n-m} \sin(n-m)x \Big|_{-\pi}^{\pi} + \frac{1}{n+m} \sin(n+m)x \Big|_{-\pi}^{\pi} \right] \\ &= 0 \end{aligned}$$

可以试着证明:

$$\int_{-\pi}^{\pi} \cos nx \sin mx dx = 0 \quad n \neq m$$

$$\int_{-\pi}^{\pi} \sin nx \sin mx dx = 0 \quad n \neq m$$



$$\begin{aligned} \vec{a} \cdot \vec{b} &= (2, 1) \cdot (-1, 2) \\ &= 2(-1) + 1 \times 2 \\ &= 0 \end{aligned}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \varphi$$

$$\vec{a} \cdot \vec{b} = 0 \quad \text{正交}$$

$$\vec{a} = (1, 2, 5)$$

$$\vec{b} = (1, 2, -1)$$

$$\vec{a} \cdot \vec{b} = 1 \cdot 1 + 2 \cdot 2 - 5 \cdot 1 = 0$$

$$\vec{a} = (a_1, a_2, a_3, \dots, a_n)$$

$$\vec{b} = (b_1, b_2, b_3, \dots, b_n)$$

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$= \sum_{i=1}^n a_i b_i = 0$$

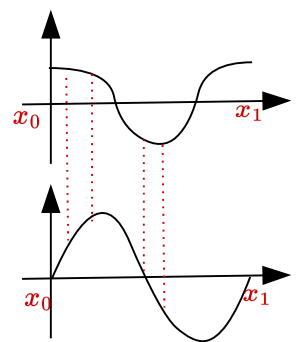
IF $m = n$?

$$\int_{-\pi}^{\pi} \cos mx \cos mx dx \quad \swarrow \frac{1}{2} [1 + \cos 2mx]$$

$$= \frac{1}{2} \left[\int_{-\pi}^{\pi} 1 dx + \int_{-\pi}^{\pi} \cos 2mx dx \right] \quad \text{cos } 0x \cdot \cos 2mx = 0$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 1 dx = \frac{1}{2} x \Big|_{-\pi}^{\pi} = \pi$$

$$a = f(x)$$



$$b = g(x)$$

$$a \cdot b = \int_{x_0}^{x_1} f(x)g(x)dx = 0$$

这两个函数正交

傅里叶级数与变换

Part2 周期为"2π"的函数展开为傅里级数

复习: 三角函数的正交性

取两个不同的, 乘积在 $[-\pi, \pi]$, 定积分为0

$$\{ \overset{\substack{\uparrow \\ \sin 0x}}{0}, \overset{\substack{\uparrow \\ \cos 0x}}{1}, \sin x, \cos x, \sin 2x, \cos 2x, \dots \sin nx, \cos nx, \dots \sin mx, \cos mx, \dots \}$$

上面集合任取两个不同的项

$n \neq m$

$n = m$

$$\int_{-\pi}^{\pi} \sin x \sin nx dx = 0$$

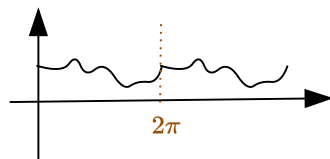
$$\int_{-\pi}^{\pi} \cos nx \cos nx dx = \pi$$

$$\int_{-\pi}^{\pi} \cos 2x dx = 0$$

$$\int_{-\pi}^{\pi} \cos nx \sin mx dx = 0$$

周期 $T = 2\pi$

$$f(x) = f(x + 2\pi)$$



像这样的函数
可以展开三角级数

教科书的形式:

$$f(x) = \sum_{n=0}^{\infty} a_n \cos nx + \sum_{n=0}^{\infty} b_n \sin nx$$

$$f(x) = \boxed{\frac{a_0}{2}} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

区别:n的起始值, 多了一项

$$= \overset{1}{a_0} \cos 0x + \sum_{n=1}^{\infty} a_n \cos nx + \overset{0}{b_0} \sin 0x + \sum_{n=1}^{\infty} b_n \sin nx$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

1. 找 a_0

$$\int_{-\pi}^{\pi} dx$$

$$\int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} a_0 dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} \overset{=0}{a_n \cos nx} dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} \overset{=0}{b_n \sin nx} dx$$

$$\int_{-\pi}^{\pi} f(x) dx = a_0 \int_{-\pi}^{\pi} dx = a_0 x \Big|_{-\pi}^{\pi} = 2\pi a_0 \quad \Rightarrow \quad a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

2.找 a_n

①等式两边乘 $\cos mx$

② $\int_{-\pi}^{\pi} dx$

$$f(x) \cos mx = \frac{a_0}{2} \cos mx + \sum_{n=1}^{\infty} a_n \cos nx \cos mx + \sum_{n=1}^{\infty} b_n \sin nx \cos mx$$

$$\int_{-\pi}^{\pi} f(x) \cos mx dx = \int_{-\pi}^{\pi} \frac{a_0}{2} \cos mx dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} a_n \cos nx \cos mx dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} b_n \sin nx \cos mx dx$$

$$\int_{-\pi}^{\pi} f(x) \cos mx dx = \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} a_n \cos nx \cos mx dx$$

$n = m$

$$\int_{-\pi}^{\pi} f(x) \cos mx dx = \int_{-\pi}^{\pi} a_n \cos nx \cos mx dx = a_n \int_{-\pi}^{\pi} (\cos nx)^2 dx = a_n \pi$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

3.找 b_n

①等式两边乘 $\sin mx$

② $\int_{-\pi}^{\pi} dx$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$f(x) = f(x + 2\pi) \quad T = 2\pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$T = 2L ?$$

$$e^{i\theta} = \cos \theta + i \sin \theta ? \quad \text{通过欧拉公式用复数的形式表示傅里叶的展开}$$

傅里叶级数与变换

Part3 周期为"2L"的函数展开为傅里级数

$f(t) = f(t + 2L)$

换元的方法

$$x = \frac{\pi}{L}t$$
$$t = \frac{L}{\pi}x$$

t	x
$2L$	2π
$4L$	4π
0	0

$f(t) = f(\frac{L}{\pi}x) \triangleq g(x)$

$g(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$

① $f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi}{L}t + b_n \sin \frac{n\pi}{L}t)$

$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} g(x)dx$

② $a_0 = \frac{1}{L} \int_{-L}^L f(t)dt$

$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \cos nxdx$

③ $a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi}{L}tdt$

$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \sin nxdx$

④ $b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi}{L}tdt$

$x = \frac{\pi}{L}t$

$\cos nx = \cos \frac{n\pi}{L}t$

$\sin nx = \sin \frac{n\pi}{L}t$

$g(x) = f(t)$

$\int_{-\pi}^{\pi} dx = \int_{-L}^L d\frac{\pi}{L}t$

$\frac{1}{\pi} \int_{-\pi}^{\pi} dx = \frac{1}{\pi} \frac{\pi}{L} \int_{-L}^L dt$

$= \frac{1}{L} \int_{-L}^L dt$

x	t
$-\pi$	$-L$
π	L

工程中: 时间没有负数, t 从0开始, 周期为 $T=2L$

$$\omega = \frac{\pi}{L} = \frac{2\pi}{T}$$

$$\int_{-L}^L dt \implies \int_0^{2L} dt \implies \int_0^T dt$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$a_0 = \frac{2}{T} \int_0^T f(t) dt$$

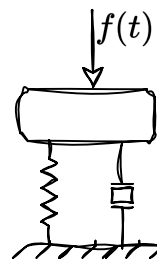
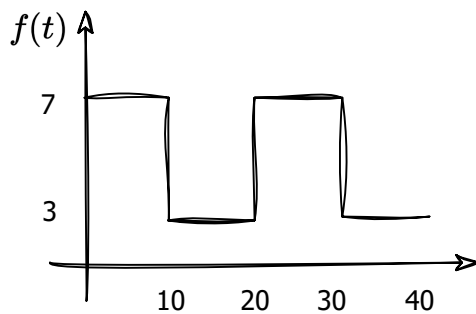
$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

$$T \rightarrow \infty$$

$f(t)$ 就不再为周期函数

Fourier transform



力施加在振动系统上

1. 分析系统的响应
2. 找到固有频率
3. 分析周期函数的展开

如果正好固有频率, 振幅过大
系统可能出问题

$$T = 20 \quad \omega = \frac{2\pi}{T} = \frac{1}{10}\pi$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$a_0 = \frac{2}{T} \int_0^T f(t) dt = \frac{1}{10} \left(\int_0^{10} 7 dt + \int_{10}^{20} 3 dt \right) = 10$$

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^T f(t) \cos n\omega t dt = \frac{1}{10} \left(\int_0^{10} 7 \cos \frac{n}{10}\pi t dt + \int_{10}^{20} 3 \cos \frac{n}{10}\pi t dt \right) \\ &= \frac{1}{10} \left(\frac{70}{n\pi} \sin \frac{n\pi t}{10} \Big|_0^{10} + \frac{30}{n\pi} \sin \frac{n\pi t}{10} \Big|_{10}^{20} \right) = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^T f(t) \sin n\omega t dt = \frac{1}{10} \left(\int_0^{10} 7 \sin \frac{n}{10}\pi t dt + \int_{10}^{20} 3 \sin \frac{n}{10}\pi t dt \right) \\ &= \frac{1}{10} \left(-\frac{70}{n\pi} \cos \frac{n\pi t}{10} \Big|_0^{10} - \frac{30}{n\pi} \cos \frac{n\pi t}{10} \Big|_{10}^{20} \right) \end{aligned}$$

n 为偶数时

$$\cos \frac{n\pi t}{10} \Big|_0^{10} = \cos \frac{n\pi t}{10} \Big|_{10}^{20} = 0$$

n 为奇数时

$$b_n = \frac{1}{10} \left(\frac{140}{n\pi} - \frac{60}{n\pi} \right) = \frac{8}{n\pi}$$

$$f(t) = 5 + \sum_{n=1}^{\infty} \frac{8}{n\pi} \sin \frac{n\pi}{10} t \quad n = 1, 3, 5, 7, \dots$$

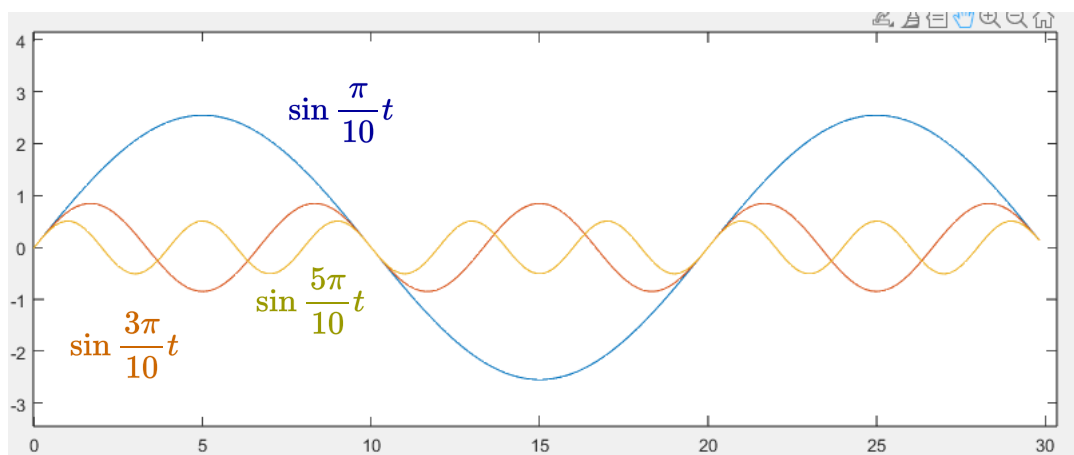
$$n \uparrow \quad \frac{8}{n\pi} \rightarrow 0$$

低频率的占了主要的部分，高频率的小的多

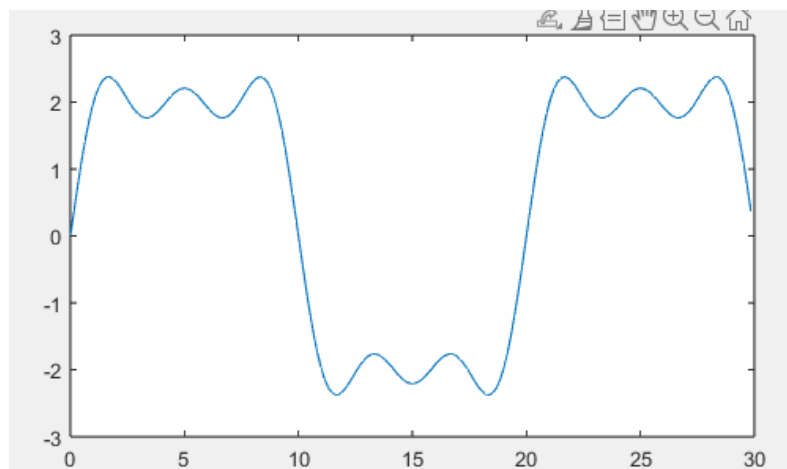
$$n = 1 \quad \frac{8}{\pi} \sin \frac{\pi}{10} t$$

$$n = 3 \quad \frac{8}{3\pi} \sin \frac{3\pi}{10} t$$

$$n = 5 \quad \frac{8}{5\pi} \sin \frac{5\pi}{10} t$$



加起来



傅里叶级数与变换

Part4 傅里叶级数的复数形式

$$f(t) = f(t + T) \quad T \text{ 周期}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t) \quad \omega = \frac{2\pi}{T}$$

$$a_0 = \frac{2}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

欧拉公式的证明

[【工程数学基础】5_如何证明宇宙第一美公式?? —欧拉公式证明](#)

[【工程数学基础】6_SinX=2? 复变函数 欧拉公式](#)

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$$

$$\sin \theta = -\frac{1}{2}i(e^{i\theta} - e^{-i\theta})$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \frac{1}{2}(e^{in\omega t} + e^{-in\omega t}) - \frac{1}{2}ib_n(e^{in\omega t} - e^{-in\omega t}) \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{a_n - ib_n}{2} e^{in\omega t} + \frac{a_n + ib_n}{2} e^{-in\omega t} \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{a_n - ib_n}{2} e^{in\omega t} + \sum_{n=1}^{\infty} \frac{a_n + ib_n}{2} e^{-in\omega t} \quad n \rightarrow (-n)$$

$$= \sum_{n=0}^0 \frac{a_0}{2} e^{in\omega t} + \sum_{n=1}^{\infty} \frac{a_n - ib_n}{2} e^{in\omega t} + \sum_{n=-\infty}^{-1} \frac{a_{-n} + ib_{-n}}{2} e^{in\omega t} \quad n \rightarrow (-\infty, \infty)$$

$$= \sum_{n=-\infty}^{\infty} C_n e^{in\omega t}$$

$$C_n = \begin{cases} \frac{a_0}{2}, & n = 0 \\ \frac{a_n - ib_n}{2}, & n = 1, 2, 3, 4 \dots \\ \frac{a_n + ib_n}{2}, & n = -1, -2, -3, -4 \dots \end{cases}$$

$$a_0 = \frac{2}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

$$C_{n=0} = \frac{a_0}{2} = \frac{1}{2} \frac{2}{T} \int_0^T f(t) dt = \boxed{\frac{1}{T} \int_0^T f(t) dt}$$

$$C_{n=1,2,3\dots} = \frac{1}{2} \left(\frac{2}{T} \int_0^T f(t) \cos n\omega t dt - i \frac{2}{T} \int_0^T f(t) \sin n\omega t dt \right)$$

$$= \frac{1}{T} \int_0^T f(t) (\cos n\omega t - i \sin n\omega t) dt$$

$$\rightarrow \cos n\omega t - i \sin n\omega t$$

$$= \cos(-n\omega t) + i \sin(-n\omega t)$$

$$= e^{-in\omega t}$$

$$= \boxed{\frac{1}{T} \int_0^T f(t) e^{-in\omega t} dt}$$

$$C_{n=-1,-2,-3\dots} = \frac{1}{2} \left(\frac{2}{T} \int_0^T f(t) \cos(-n)\omega t dt + i \frac{2}{T} \int_0^T f(t) \sin(-n)\omega t dt \right)$$

$$= \frac{1}{T} \int_0^T f(t) (\cos n\omega t - i \sin n\omega t) dt = \boxed{\frac{1}{T} \int_0^T f(t) e^{-in\omega t} dt}$$

$$n = 0: \quad \frac{1}{T} \int_0^T f(t) e^0 dt = \boxed{\frac{1}{T} \int_0^T f(t) dt}$$

$$f(t) = f(t + T)$$

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega t}$$

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-in\omega t} dt$$

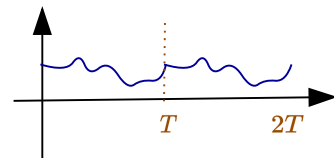
傅里叶级数与变换

Part5 傅里叶变换 FT

$$f_T(t) = f(t + T)$$

定义了函数

$$f_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t} \quad (1) \quad \omega_0 = \frac{2\pi}{T} \text{ 基频率}$$



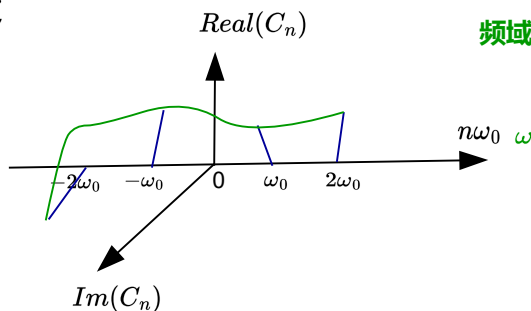
时域表达



频谱

$|C_n|$ 幅度

$$C_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f_T(t) e^{-in\omega_0 t} dt \quad (2) \quad C_n = a + bi$$



频域表达

非周期函数的一般形式

无限久后重复 $T \rightarrow \infty$

$$\lim_{T \rightarrow \infty} f_T(t) = f(t)$$

两个频率之间的距离定义为

$$\dots + C_{-1}e^{-i\omega_0 t} + C_0e^0 + C_1e^{i\omega_0 t} + C_2e^{i2\omega_0 t} + \dots$$

离散 $\rightarrow$ 连续

$$\Delta\omega = (n+1)\omega_0 - n\omega_0 = \omega_0 = \frac{2\pi}{T}$$

$T \uparrow, \Delta\omega \downarrow$ 距离就没有了, 连续了

把(2)代入(1)

$$f_T(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f_T(t) e^{-in\omega_0 t} dt e^{in\omega_0 t}$$

$$f_T(t) = \sum_{n=-\infty}^{\infty} \frac{\Delta\omega}{2\pi} \int_{-\frac{T}{2}}^{\frac{T}{2}} f_T(t) e^{-in\omega_0 t} dt e^{in\omega_0 t}$$

$$\frac{1}{T} = \frac{\Delta\omega}{2\pi}$$

$T \rightarrow \infty$

$$\int_{-\frac{T}{2}}^{\frac{T}{2}} dt \rightarrow \int_{-\infty}^{+\infty} dt$$

$$n\omega_0 \rightarrow \omega$$

$$\sum_{n=-\infty}^{\infty} \Delta\omega \rightarrow \int_{-\infty}^{+\infty} d\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt e^{i\omega t} d\omega$$

FT
傅里叶变换

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt$$

逆变换

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega$$

傅里叶级数与变换

知识点

① 三角函数的正交性

取两个不同的, 乘积在 $[-\pi, \pi]$, 定积分为0

$$\{0, 1, \sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin nx, \cos nx, \dots\}$$

$\sin 0x$ $\cos 0x$ $\sin nx$ $\cos nx$ $n = 0, 1, 2, 3, \dots$

正交: 上面集合任取两个不同的项

$$\int_{-\pi}^{\pi} \cos x \cos 2x dx = 0$$

$$\int_{-\pi}^{\pi} \sin nx dx = 0$$

② 周期为' 2π '

$$f(x) = f(x + 2\pi)$$

$$f(x) = \sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\int_{-\pi}^{\pi} dx$$

$$\int_{-\pi}^{\pi} \cos nx dx$$

$$\int_{-\pi}^{\pi} \sin nx dx$$

左边和右边同时

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

③ 周期为' $2L$ '

$$T = 2L \quad f(t) = f(t + 2L)$$

$$x = \frac{\pi}{L} t$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi}{L} t + b_n \sin \frac{n\pi}{L} t)$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi}{L} t dt$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi}{L} t dt$$

④ 复指数

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$f(t) = f(t + T) \quad \omega = \frac{2\pi}{T}$$

$$f(t) = \sum_{-\infty}^{\infty} C_n e^{in\omega t}$$

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-in\omega t} dt$$

⑤ TF

$$f(t) = f(t + T) \quad T \rightarrow \infty$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt e^{i\omega t} d\omega$$

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt \quad \text{FT}$$

$$s = i\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega \quad \text{IFT}$$

$$F(s) = \int_{-\infty}^{+\infty} f(t) e^{-st} dt \quad \text{Laplace}$$

desmos <https://www.desmos.com/calculator?lang=zh-CN>